The Design of a Multi-bit Sigma-delta ADC Modulator

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***Abstract*—Sigma-delta modulation is a superior technique to realize A/D converter with high resolution. This paper proposes a new method to design a low-cost, high-performance and third-order sigma-delta modulator whose structure is three-bit quantization CIFF (cascade of integrators, feed forward form). It combines the high-speed and data processing ability of FPGA to achieve high accuracy of sigma-delta ADC. The diagram of each part in circuit is given, and system modeling and simulations are carried out in this paper. The results show that, the modulator with multi-bit quantization has a higher SNR.**

Keywords—Sigma-delta ADC; modulator; over sampling; noise shaping; multi-bit quantization

I. INTRODUCTION

With the progress of the technology, people pay more and more attention on the data conversion accuracy. For example, it needs the analog to data converter with high accuracy in the hi gh fidelity audio system, namely, AD converter must have mor e than 16bit resolution. However, when adapting the traditional AD convert principle based on the double integral model or th e successive approximating model, it won't mean the so high a ccuracy requirement.

Sigma-delta is an implementation method to realize converter with high accuracy. It uses the over sampling principle to compress the energy of quantization noise largely in the signal frequency band, and finally, through the down-sampling accomplished by digital decimation filter, so the higher SNR can be got. Comparing with the converters with other structures, sigma-delta ADC has the advantages of high accuracy, high linearity, large dynamic range and so on.

Sigma-delta modulator mainly consists of modulator and digital decimation filter, and according to bit of the modulator, it can be divided into single-bit quantization and multi-bit quantization. The current low order and single-bit ADC [1] is limited by the order of modulator and single-bit quantization, which decreases the possibility to get the higher SNR. With the wider application of sigma-delta ADC, the requirements to it become higher. The SNR is the uppermost parameter to measure performance, and it can be improved largely by changing the modulator order, over sampling rate and quantization bits. Even though increasing the modulator order can largely improve the SNR, the attendant problems have emerged, such as the stability and complexity of circuit, and so on [2]. These problems are also serious in low voltage and low power circuits at present. Therefore, the multi-bit sigma-delta modulator will be the market mainstream because of the performance advantage and wide application prospect [3].

II. MULTI-BIT QUANTIZATION SIGMA-DELTA MODULATOR TRANSFORMATION TECHNIQUE

The differences between sigma-delta ADC and general Nyquist converter are that the former adapts the over-sampling and noise shaping technique. Based on these two techniques, the paper designs a high performance sigma-delta ADC modulator. The transformation technologies adopted are as followed.

A. Over-sampling Technique

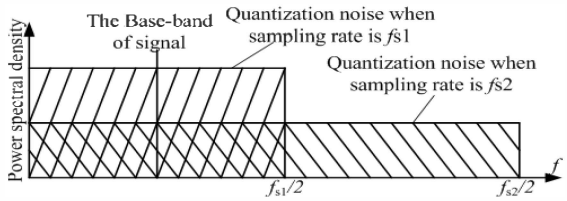

Fig. 1. Quantization noise distribution in different sampling frequency

Over sampling is defined as sampling the analog signal using the frequency which is much higher than the Nyquist frequency. In general, the over sampling technique uses over-sampling ratio (OSR) to measure the over sampling level. There are two advantages when adapting the over sampling: first, high sampling rate can reduce the design specification of preceding anti-alias analog filter. Even if the stop band attenuation of anti-alias filter is not enough around the signal cut-off frequency, it can't produce serious aliasing of signal, and the distortion won't be very high after the recovery. Second, over sampling technique can increase signal quantization to noise ratio (SQNR). According to the signal sampling quantization theory, if the smallest amplitude of the input signal is greater than the quantization step Δ, and the amplitude distributes randomly, then the total power of quantization noise is a constant, and it has nothing to do with sampling frequency f_s, in addition, it distributes uniformly in $0\sim f_s/2$. Therefore, the quantization noise level is inversely proportional to the sampling frequency, and increasing the sampling frequency can reduce the quantization noise level,

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while the base-band is constant. So it reduces the noise function in the base-band to improve the SNR. Figure 1 shows the distribution of quantization noise at different frequency, and it clearly describes the relationship between sampling frequency and noise level. Here, f_{s2} is far greater than f_{s1}, so s quantization noise power is much smaller than f_{s1}'s in the base-band [4].

B. Noise Shaping Technique

The basic principle of noise shaping technique is that shaping the quantization noise distributing in [$-f_s/2$, $+f_s/2$] uniformly. Noise shaping doesn't reduce the total quantization noise power, but transfers the low frequency noise power to the high frequency to make most quantization noise power out of the bandwidth that we wanted, thus improves the SNR further. Sigma-delta modulator can shape the noise without influencing the signal bandwidth, and can combine the noise shaping with over sampling well [5]. Figure 2 is the basic structure of one order Sigma-delta ADC.

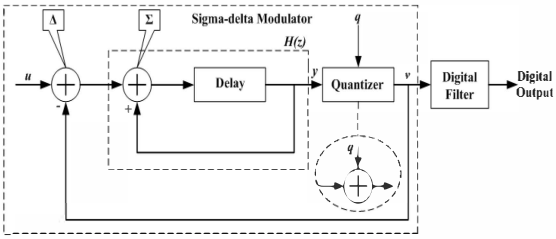

Fig. 2. One order Sigma-delta ADC schematic diagram

It assumes that the quantization noise distributes uniformly, and it is independent of the input signal u. So the quantizer can be abstracted as a simple adder model, and it is marked in Figure 2 by the elliptical dotted line frame. This linear system simplified from modulator has two inputs u、q and one output v, the signal transfer function from u to v and noise transfer function from q to v are as followed relatively:

$$STF(z)=\frac{V(z)}{U(z)}=\frac{H(z)}{1+H(z)}=z^{-1} \tag{1}$$

$$NTF(z)=\frac{V(z)}{Q(z)}=\frac{1}{1+H(z)}=1-z^{-1} \tag{2}$$

These two functions show that, the input signal u appears steadily at the output after one clock delay, and the delayed signal adding quantization noise component shaped by one order high-pass will become the output v. The order of noise shaping is relative to the order of noise transfer function [6].

C. Multi-bit Quantizer Technique

To improve the single sigma-delta modulator performance largely, the multiple bits quantizer technique is needed. Sigma-delta modulator with multi-bit quantization structure can increase the convert rate and the resolution of ADC, and it consists of an N-bit parallel ADC and an N-bit DAC. Compared with the single bit quantization modulator, the multi-bit one has many advantages when the condition is same: it has a bigger stable region, a larger input dynamic range, a higher resolution and most importantly, it has a higher linearity.

Quantizer is the only nonlinear component in Sigma-delta modulator, but multi-bit quantizer has a good stability and a high linearity, so its performance is close to the linear system [7].

In sigma-delta modulator circuit, the structure of single bit quantizer is simple. It just needs one comparator and the DAC in the feedback circuit has a stable linearity. However, single bit quantizer has a large quantization noise, and needs the higher OSR to suppress it. Using the single bit quantizer in high order single circuit modulator will make the system become instable. As for the multi-bit quantization, it has higher conversion accuracy, reduces the quantization noise, eliminates the relativity between modulation process and input signal effectively, and enhances the stability of system. While, multi-bit quantization needs adding a multi-bit DAC to the feedback circuit to produce the feedback signal, which will lead into the nonlinear problem and affect the system's performance, so the calibration is necessary. The quantization SNR of an L-order and M-bit sigma-delta modulator is as followed [8].

$$\begin{aligned} SQNR &= 10\log_{10}\left(\frac{P_s}{P_Q}\right)=10\log_{10}\left(\frac{3}{2}2^{2N}\right) \\ &+10\log_{10}\left(\frac{2L+1}{\pi^{2L}}OSR^{2L+1}\right) \\ &+10\log\left[\left(2^M-1\right)^2\right] \end{aligned} \tag{3}$$

Here, M presents the digit of quantizer. The relationship between the digit and SNR is shown in Table I.

TABLE I. RELATIONSHIP BETWEEN THE BIT OF QUANTIZER AND SNR

The Bit of Quantizer	The SNR Increment	The Increment of SNR When Quantizer Increases One Bit
2	9.5	9.5
3	17	7.5
4	23.5	6.5
5	29.8	6.3
6	35.9	6.2

From Table I, we can see that increasing the digit of quantizer can increase SNR quickly. When the digit of quantizer is more than 5, the quantization accuracy in feedback circuit becomes double, and the increment of SNR is about 6 dB [9].

III. DESIGN OF MULTI-BIT QUANTIZATION SIGMA-DELTA MODULATOR

Sigma-delta modulator is the core part of sigma-delta ADC, and it is composed of the difference summation unit, integrator, quantizer and a DAC. The paper designs a sigma-delta modulator with 24 significant digits, uses Simulink toolbox fully in MATLAB to model the multi-bit quantization sigma-

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delta modulator, gets its topological structure, and finally designs each module circuit of modulator by using simple components.

A. Sigma-delta Modulator Modeling

The design adapts single circuit three orders CIFF circuit, and the ideal model of multi-bit quantization sigma-delta modulator is established by Simulink in MATLAB. It is shown in Figure 3.

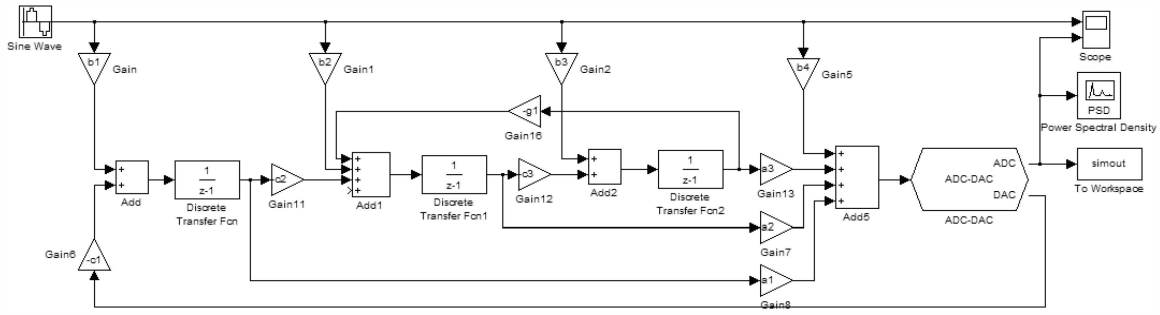

Fig. 3. Multi-bit quantization sigma-delta modulator's ideal model

B. Sigma-delta Modulator Circuit Design

Sigma-delta modulator includes integrator, quantizer and DAC. The paper designs a low cost and high performance sigma-delta modulator through simple discrete device. It adapts 3-bit quantizer and feedback DAC. Compared with others, they can be achieved easily and their cost is low. The each part circuit is as followed.

1) Integrator design

The integrator is the core circuit module of sigma-delta modulator. Here we adapt the switched capacitor integrator, which consists of the switch, sampling integrating capacitor and operational amplifier. Figure 4 describes the three stages integrator with switched capacitor. A voltage follower between input port and first stage integrator is designed to lead into the negative feedback with voltage series. The input impedance of voltage follower is high, while the output impedance is low, so it can achieve the impedance match and isolation between former and latter stages.

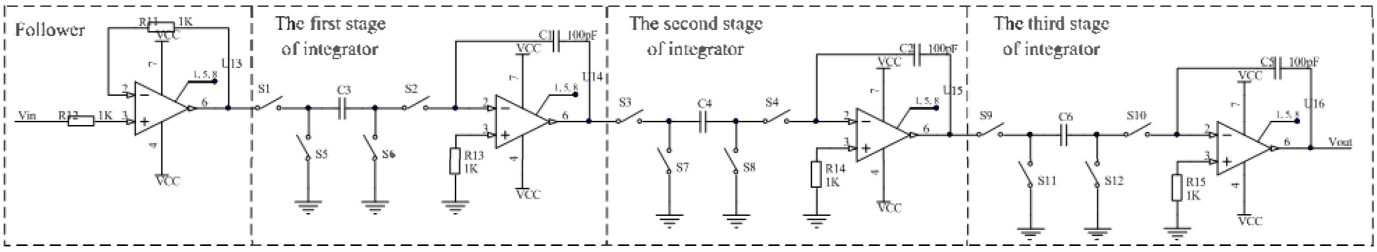

Fig. 4. Three orders paralleled integrator

2) DAC design

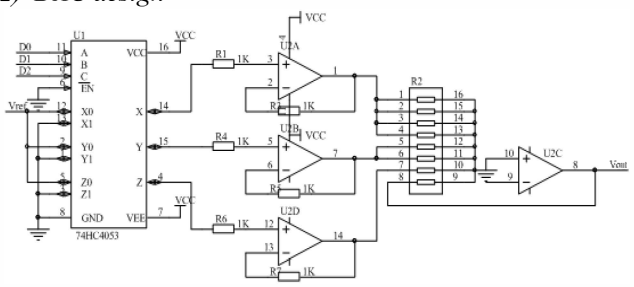

Fig. 5. Feedback ADC

The output of quantizer gets to the first integrator through the feedback of DAC. The feedback loop can make the input port of the first integrator tend to zero, namely, the input value of modulator is equal to the average value of DAC's output. In this paper, we use 3-bit feedback ADC, which is composed of multiple switch 74HC4053, operational amplifier, and exclusions. Due to the low cost and fast speed of multiple switch, and similar performance in exclusion, they are appropriate to used as feedback ADC whose accuracy is very high. Its structure is shown in Figure 5.

3) Quantizer design

The quantizer is a 3-bit successive approximation ADC, and is composed of a comparator, D/A converters, buffer registers and control logic circuits. As shown in Figure 5, the control logic is realized in FPGA. The basic principle is that, comparing levels from high bit to low bit, as if using balance to weigh objects, we should increase or decrease weights step by step from heavy to light. The process of successive approximation conversion is as follows: clear every bit when initializing, and the supreme bit will be 1 and be sent into D/A convert at the beginning of conversion. The output of D/A convert is analog, i.e., *Vo*. Then compare it to *Vi*, which is also an analog signal waiting to be converted. If *Vo*<*Vi*, the 1 in this bit is kept, otherwise it will be cleared. Repeat this process

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until the lowest bit of the successive approximation register. After the transformation, sent every digital quantity into the buffer register, and get the digital output. All these processes are controlled by a circuit. Because the 3-bit successive approximation ADC can be achieved by sample discrete components, and doesn't need photolithographic process, thus it has low cost and high convert rate.

IV. Simulation Experiment Result

At different over-sampling rates, we simulate the proposed modulator which has a monocyclic third-order CIFF structure with 3-bit quantization, and the simulation results are shown in Table II.

TABLE II. THE RELATIONSHIP BETWEEN OVER-SAMPLING RATE AND MODULATOR PERFORMANCE

Over Sampling Rate	SNR	Number of Significant Bit
16	82.0	13.32
32	102.5	16.74
64	123.7	20.25
128	144.9	23.70
256	169.1	27.79
512	189.9	31.26

From the table, we can see that, when the number of order and bits of quantization in the modulator are determined, increasing the sampling rate can obviously improve the modulator performance.

Thus, making full use of the high speed and data processing ability of FPGA, it is easy to realize high over-sampling rate and sigma delta ADC with high performance.

The paper simulates the 3-bit quantization sigma-delta modulator under the conditions that the over-sampling rate is 256. Here we use the Simulink toolbox in MATLAB. Figure 6 shows the performance of multi-bit quantization modulator model while figure 7 shows the performance of modulator model with 1-bit quantization.

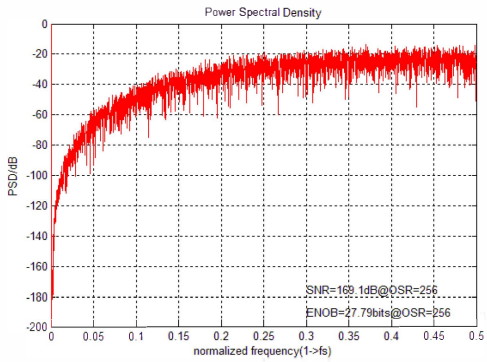

Fig. 6. The performance of multi-bit quantization modulator model

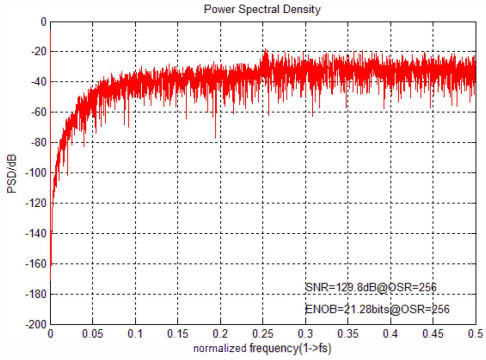

Fig. 7. The performance of single bit quantization modulator model

The simulation results show that, the wave groove in Figure 6 is wider than it in Figure 7, and the SNR is evidently higher than it by one bit quantization. when sampling rate is 256, the resolution of 3-order 3-bit quantization sigma delta modulator is 27.79 bits when the SNR is 169.1dB, while the resolution of 1-bit quantization sigma-delta modulator is 21.28 bits when SNR is 129.8 dB.

V. Conclusion

Based on the study of sigma-delta ADC's speed and accuracy, this paper completes its system modeling, simulation and obtains the topology structure by using the Simulink tool in MATLAB. A multi-bit quantization sigma-delta modulator with simple circuit is designed by discrete components. Combined with digital decimation filter realized by FPGA, the modulator can achieve high accuracy sigma-delta ADC.

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